

Code-Layout, Readability and Reusability

Date	04 july 2026
TeamID	
ProjectName	A Comprehensive Measure of Well-Being
MaximumMarks	5Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the code base in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/ Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation is maintained for better readability.
2	Proper File Structure	Files and folders are logically organized	Yes	Project follows a structured folder hierarchy (templates, static, dataset, model, etc.).
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names are used for better understanding.
4	Function / Method Names	Functions are descriptively named	Yes	Function names clearly indicate their functionality.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are added to explain important sections of the code.
6	Modular Design	Code is split into reusable functions/modules	Yes	Different tasks are divided into separate functions and modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unnecessary and duplicate code has been eliminated.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic error handling is implemented; additional validation can be added for robustness.

ReusableComponents/Modules:

S.No	Component / Module Name	Language / Technology Where Reused	Reusability Level (High/Medium/Low)
1	Data Preprocessing Module	Python (Pandas, NumPy)	High
2	Data Visualization Module	Python (Matplotlib, Seaborn)	High
3	Machine Learning Model Training Module	Python (Scikit-learn)	High
4	Prediction Module	Python, Flask	High
5	Web Interface (HTML Templates)	HTML, CSS, Flask	Medium

Overall CodeQualityAssessment:

Aspect	Rating (1-5)	Comments
Code Layout &Structure	5	The project follows a well-organized folder structure with separate files for datasets, models, templates, and application code.
Readability	5	The code uses meaningful variable names, consistent formatting, and proper indentation, making it easy to understand.
Reusability	5	Functions and modules are reusable across different parts of the project, with scope for further modularization.
Documentation / Comments	5	Important sections of the code are documented with comments, though additional comments could improve clarity.
Overall Score	4.5/5	The codebase is well-structured, readable, maintainable, and suitable for future enhancements.